Similarity and normalisation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable similarity and normalization record.

Trace norms,compare metrics,retain policies and explain the removed magnitude.

Keep one habit: W1–W3/W5 → selected new vectors → project comparison ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Last week, a neighbourhood depended on a chosen distance rule. This week, compare the same fictional vectors in two ways. Squared distance rewards nearby coordinates.

  Cosine similarity compares direction and ignores positive changes in length. Neither rule is automatically right for every task.

- For Q, length = square root of (3 squared + 4 squared) = square root of 25 = 5. Divide both components by 5 to get its **unit vector** [0.6, 0.8]. Its length is one because 0.6 squared + 0.8 squared = 1.

  A becomes the same **unit vector**. C becomes [-0.6, -0.8], pointing in the opposite direction.

- These supplied coordinates are for arithmetic practice. They are not learned language embeddings. Normalising each row keeps its direction but removes its original length unless that length is saved separately.

Embeddings and neighbourhoods

> **My remembered idea, example and correction**
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Augmentation and leakage

> **My remembered idea, example and correction**

Convolution as a local operation

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Vector	Coordinates	Length
Query Q	[3, 4]	5
Candidate A	[6, 8]	10
Candidate B	[3, 0]	3
Candidate C	[-3, -4]	5

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main query Q=[3, 4] has norm 5. A=[6, 8] has norm 10; B=[3, 0] has norm 3; C=[-3, -4] has norm 5. Unit Q and A are[.6, .8], unit B is[1, 0], unit C is[-.6, -.8]. Raw squared distances from Q to A/B/C are 25/16/100, so B wins. Cosines1/.6/-1 select A. Squared unit distances 0/.8/4 also select A and satisfy 2-2 cosine.

  Additional D=[0, 5] has raw squared distance 10, cosine.8 and unit squared distance.4; adding it changes the candidate set. Normalize both sides and retain original lengths if magnitude matters. If these vectors represent event counts, Q and A have totals 7 and14 despite matching direction. Reject a zero query for cosine; exclude/report zero candidates; return no comparison if none remain.

  Nonzero perpendicular[-4, 3]has valid cosine 0. Per-vector normalization fits no dataset statistics. Separate picture query[2, 0], P=[8, 0], candidate named Q=[2, 1]:raw squared distances 36/1 prefer candidate Q, cosines $1/(2/\sqrt{5})$prefer P. Zero P makes its cosine undefined. Extended ranking uses absolute tie tolerance 1e-9. These coordinates alone provide no task-quality or semantic-equality evidence.

- For Q [3, 4], A [6, 8] and C [-3, -4], give each length and **unit vector**. Explain one distinction that normalisation removes and one that it preserves.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main query Q=[3, 4] has norm 5. A=[6, 8] has norm 10; B=[3, 0] has norm 3; C=[-3, -4] has norm 5. Unit Q and A are[.6, .8], unit B is[1, 0], unit C is[-.6, -.8]. Raw squared distances from Q to A/B/C are 25/16/100, so B wins. Cosines1/.6/-1 select A. Squared unit distances 0/.8/4 also select A and satisfy 2-2 cosine.

  Additional D=[0, 5] has raw squared distance 10, cosine.8 and unit squared distance.4; adding it changes the candidate set. Normalize both sides and retain original lengths if magnitude matters. If these vectors represent event counts, Q and A have totals 7 and14 despite matching direction. Reject a zero query for cosine; exclude/report zero candidates; return no comparison if none remain.

  Nonzero perpendicular[-4, 3]has valid cosine 0. Per-vector normalization fits no dataset statistics. Separate picture query[2, 0], P=[8, 0], candidate named Q=[2, 1]:raw squared distances 36/1 prefer candidate Q, cosines $1/(2/\sqrt{5})$prefer P. Zero P makes its cosine undefined. Extended ranking uses absolute tie tolerance 1e-9. These coordinates alone provide no task-quality or semantic-equality evidence.

- For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule. Calculate squared distances after normalising both sides and check the relationship to cosine.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main query Q=[3, 4] has norm 5. A=[6, 8] has norm 10; B=[3, 0] has norm 3; C=[-3, -4] has norm 5. Unit Q and A are[.6, .8], unit B is[1, 0], unit C is[-.6, -.8]. Raw squared distances from Q to A/B/C are 25/16/100, so B wins. Cosines1/.6/-1 select A. Squared unit distances 0/.8/4 also select A and satisfy 2-2 cosine.

  Additional D=[0, 5] has raw squared distance 10, cosine.8 and unit squared distance.4; adding it changes the candidate set. Normalize both sides and retain original lengths if magnitude matters. If these vectors represent event counts, Q and A have totals 7 and14 despite matching direction. Reject a zero query for cosine; exclude/report zero candidates; return no comparison if none remain.

  Nonzero perpendicular[-4, 3]has valid cosine 0. Per-vector normalization fits no dataset statistics. Separate picture query[2, 0], P=[8, 0], candidate named Q=[2, 1]:raw squared distances 36/1 prefer candidate Q, cosines $1/(2/\sqrt{5})$prefer P. Zero P makes its cosine undefined. Extended ranking uses absolute tie tolerance 1e-9. These coordinates alone provide no task-quality or semantic-equality evidence.

- A candidate set contains Z [0, 0] and B [3, 0]. With nonzero query Q, what happens under this lesson’s zero-vector policy? What if Q is zero, or Z is the only candidate?

  Explain why replacing an undefined cosine with zero is misleading.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main query Q=[3, 4] has norm 5. A=[6, 8] has norm 10; B=[3, 0] has norm 3; C=[-3, -4] has norm 5. Unit Q and A are[.6, .8], unit B is[1, 0], unit C is[-.6, -.8]. Raw squared distances from Q to A/B/C are 25/16/100, so B wins. Cosines1/.6/-1 select A. Squared unit distances 0/.8/4 also select A and satisfy 2-2 cosine.

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- For event-count vectors [3, 4] and [6, 8], what useful information can normalisation remove? A team chooses cosine because it improves results on the final evaluation set, then calls that same set independent **evidence**.

  Explain how to repair the evaluation plan.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates.

  Explain the changed ranking.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Compare a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled similarity and normalisation **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate **vector length** and normalise a nonzero vector.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Compare the rankings for Similarity and normalisation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Similarity and normalisation. Proposed explanation: "Cosine is a probability". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates.

  Explain the changed ranking.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Compare angular alignment with raw distance on the same vectors.

- Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

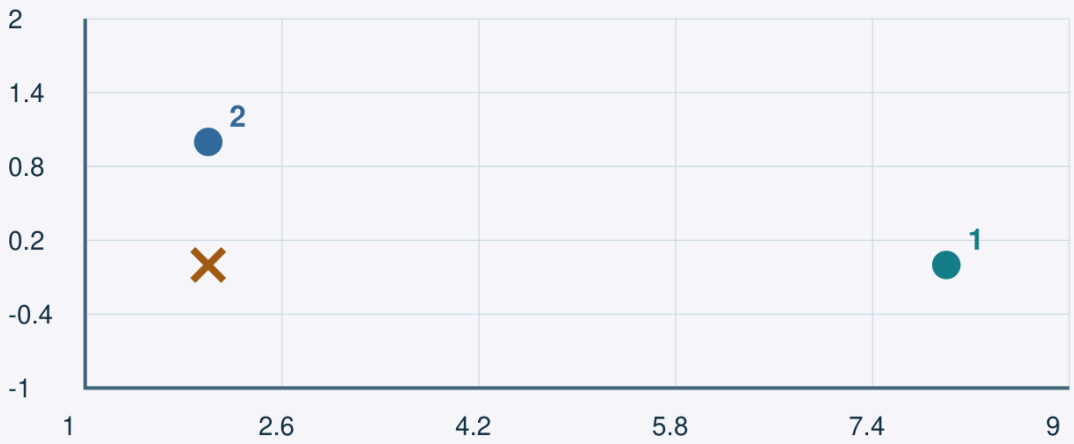
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/27

Raw proximity differs from angular alignment



Horizontal: first vector coordinate; vertical: second vector coordinate

1: P at (8, 0)

2: Q at (2, 1)

Cross: query at (2, 0)

Distance rule: Compare squared Euclidean distance and cosine

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/27

Similarity and normalisation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Raw distances

P distance 36; Q distance 1, so Q is closer in raw coordinates.



2. Angular similarity

Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely.



3. Normalise

Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

CHANGE ONE THING

Replace P with [0, 0].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compare angular alignment with raw distance on the same vectors.
- Which labelled step matches this detail? Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.
- 1. Angular similarity
- 2. Normalise
- 3. Raw distances
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Normalise Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0. Raw Euclidean distance includes differences in both direction and magnitude.

Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates. Declare which question your similarity measure answers.

A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

Changed-case question and optional hint

- Changed-case practice: Replace P with [0, 0]. Which conclusion is supported by the supplied facts?
- 1. Cosine and unit normalisation for P are undefined.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Cosine and unit normalisation for P are undefined. The zero vector has norm 0, so dividing by its norm is invalid.

Repair and retain the record

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Similarity and normalisation: the case

AI ACADEMY / AI-5/27 · 01 · LOOK AT THE FACTS

Similarity and normalisation: the case

The situation

Compare angular alignment with raw distance on the same vectors.

Supplied evidence

Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain similarity and normalisation using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

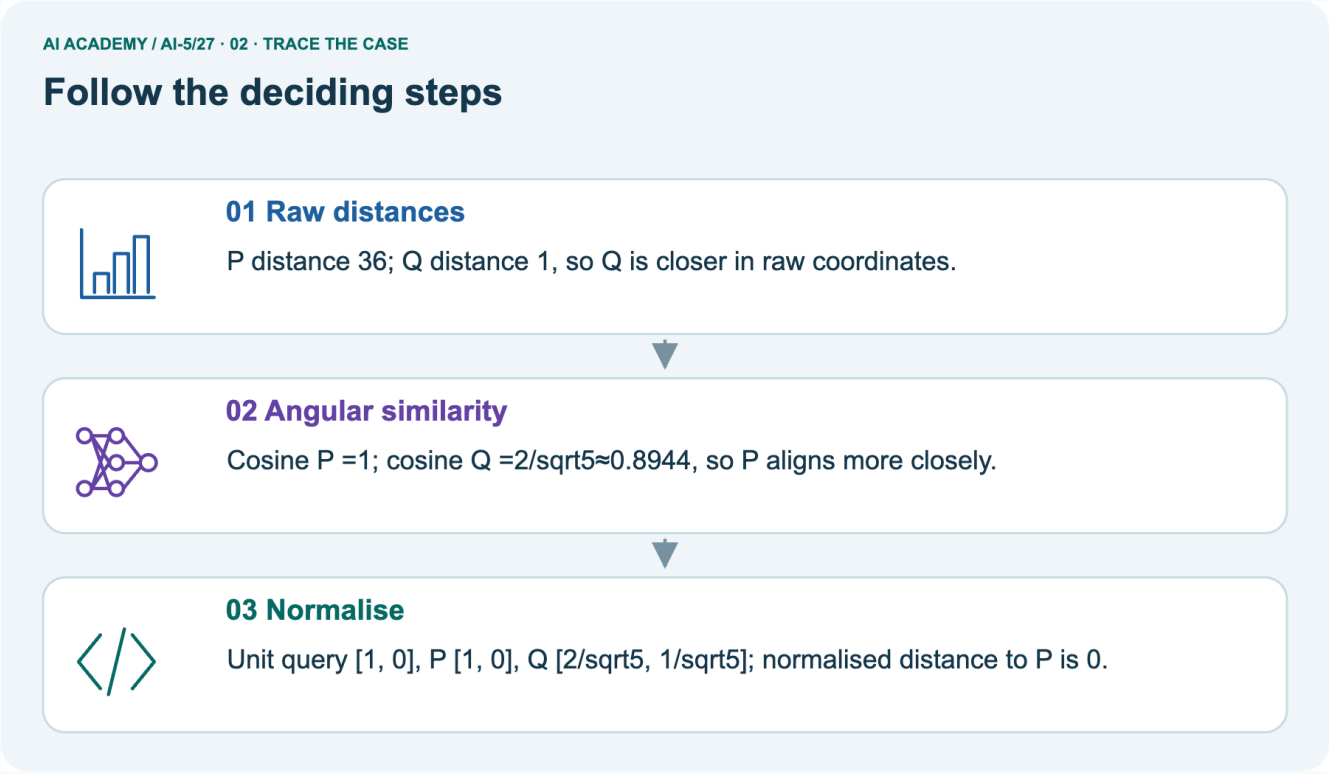
Compare angular alignment with raw distance on the same vectors. Supplied facts: Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



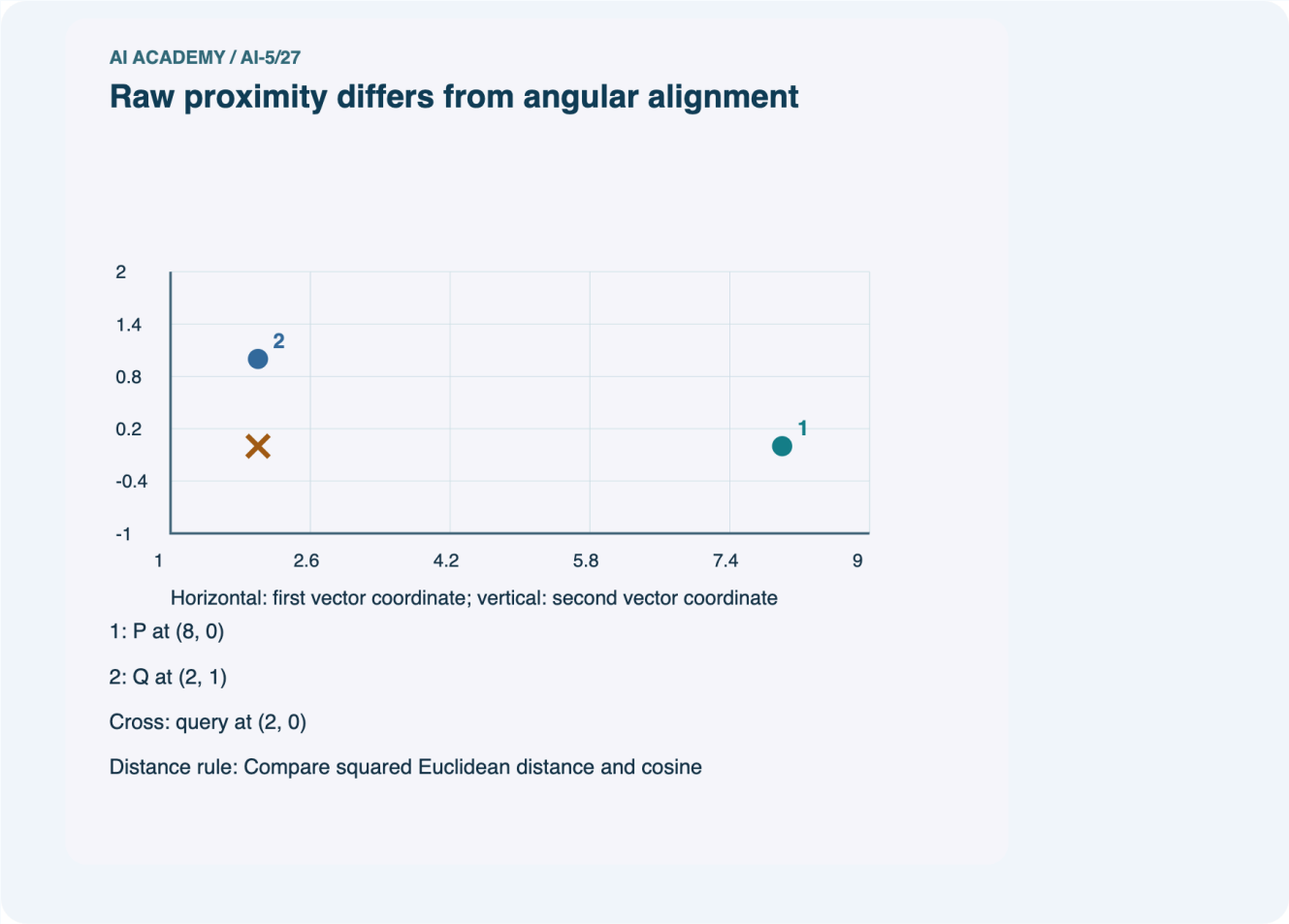
Follow the labelled picture with the matching explanation. Open the vector diagram

Raw distances: P distance 36; Q distance 1, so Q is closer in raw coordinates. → Angular similarity: Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely. → Normalise: Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Raw proximity differs from angular alignment



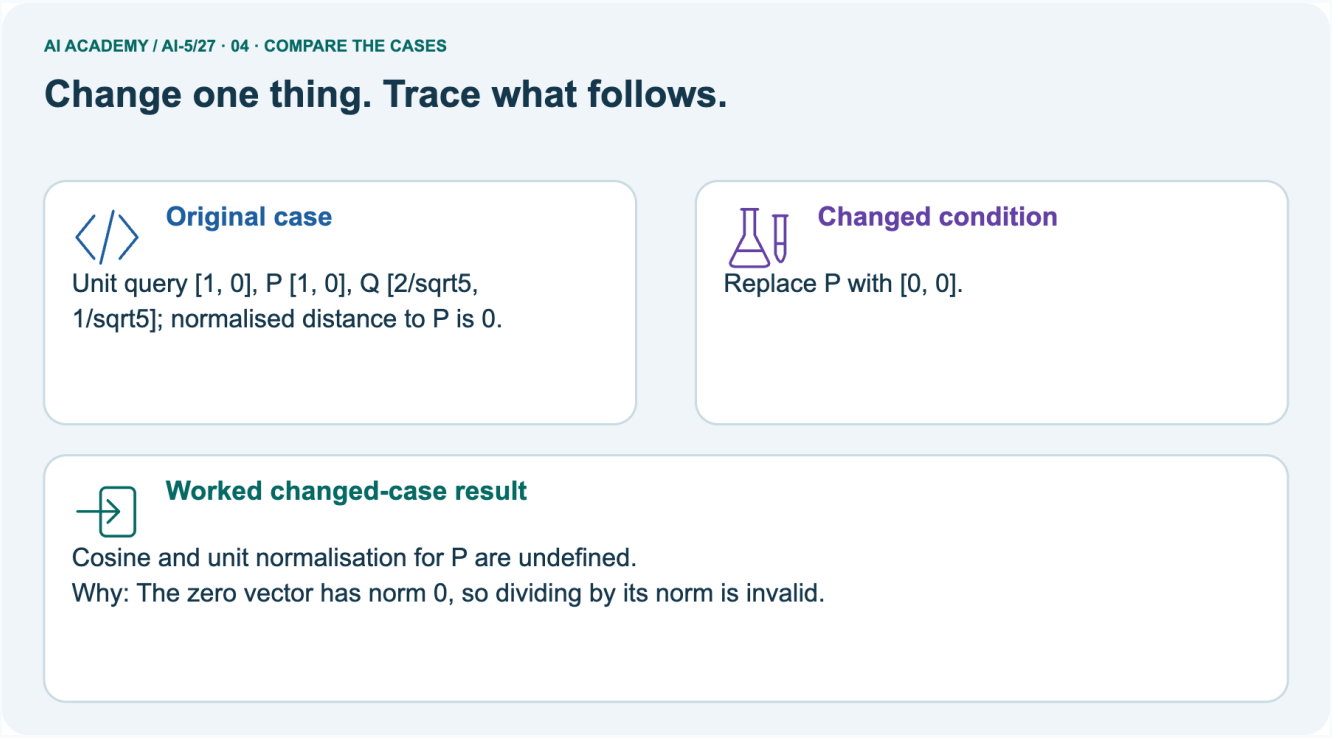
Follow the labelled picture with the matching explanation. Open the vector diagram

Raw proximity differs from angular alignment. P: (8, 0). Q: (2, 1). Query query: (2, 0). Axes: first vector coordinate; second vector coordinate. Distance: Compare squared Euclidean distance and cosine.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

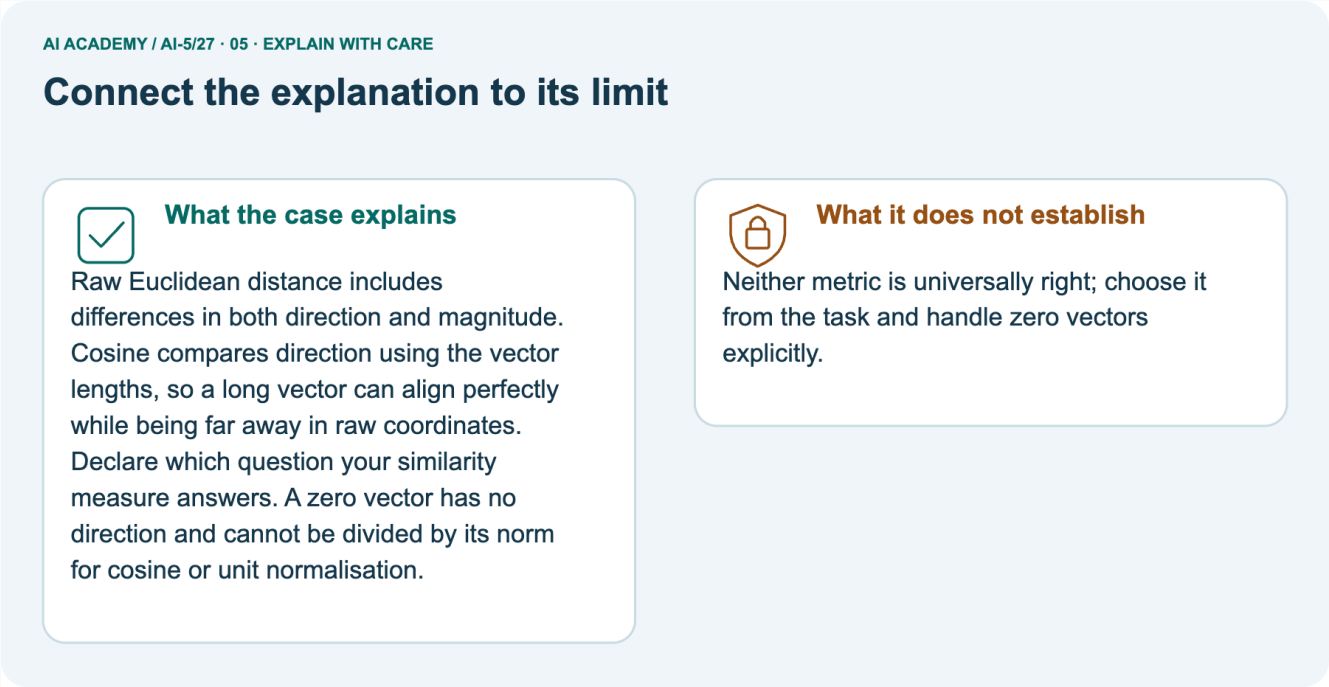
Original case: Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0. Changed condition: Replace P with [0, 0]. Expected result: Cosine and unit normalisation for P are undefined.

Reason: The zero vector has norm 0, so dividing by its norm is invalid.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Raw Euclidean distance includes differences in both direction and magnitude. Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates.

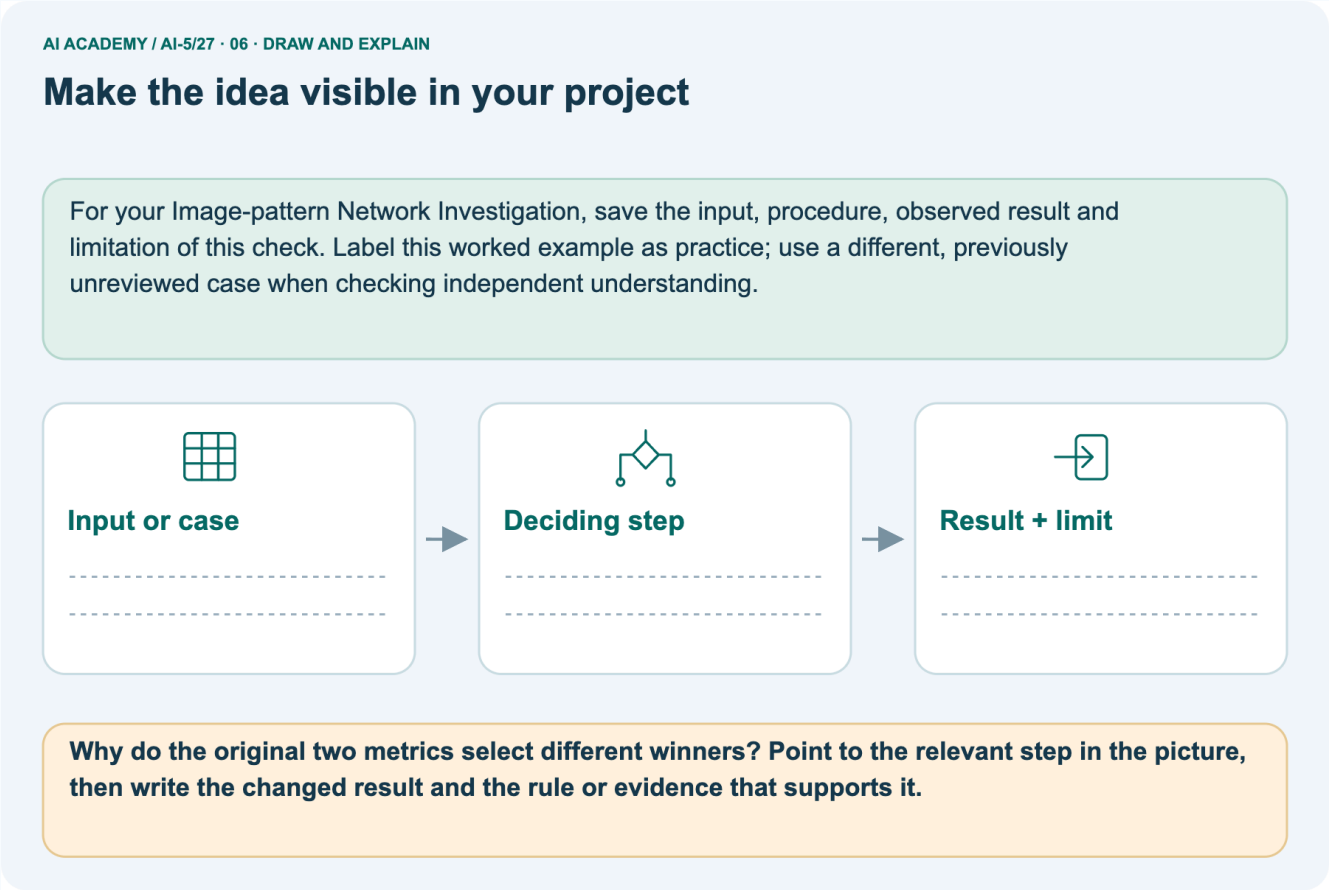
Declare which question your similarity measure answers. A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

Limit: Neither metric is universally right; choose it from the task and handle zero vectors explicitly.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Similarity and normalisation: the case

02. Follow the deciding steps

03. Raw proximity differs from angular alignment

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.